

STIs and reproductive function

How can an infection interfere with reproduction even when hormone production continues?

Learning goal

Explain how sexually transmitted infections can affect tissue, transport and fertility, and distinguish infection prevention from conception control.

Before you begin

An open, functioning reproductive pathway matters as well as gamete production and hormonal signalling.

How to complete this lesson Learn → worked example → practise → lesson check

- Read the explanation and its diagrams. Open a bold term when you need a definition.
- Follow the worked example, then try both supported questions. Select Check answer for feedback.
- Open the required check and select Start. Correct both selections to unlock writing.
- Write each explanation and select its Save written response button. Editing saved writing makes it a draft again.
- Select Submit and finish to complete the check. Find your records in All My Work.

Expected work: one completed lesson check. Supported practice, videos and textbook questions are optional. Saved writing is not automatically graded.

Chapter 14 · printed pp. 486–491

Open p. 486 · Open p. 491

Key terms: STI · pathogen · pelvic inflammatory disease · infertility

Vocabulary help

Key terms

Words for this lesson

Use these definitions when you need help with a term.

STI

A sexually transmitted infection.

pathogen

An organism or infectious agent capable of causing disease.

pelvic inflammatory disease

Inflammation involving upper female reproductive structures, often associated with infection.

infertility

Difficulty establishing a pregnancy due to one or more reproductive factors.

More vocabulary for this lesson

Lesson vocabulary

ectopic pregnancy

A pregnancy implanted outside the normal uterine site.

Identify the infectious agent before explaining its effect

A reproductive system needs functioning cells, hormone signals and open transport pathways. An infection can disturb one part while another still works. Our question is therefore not just “Are hormones present?” It is “Which tissue is affected, and what does that tissue normally do?”

A sexually transmitted infection, or STI, is an infection that can pass through sexual contact. A pathogen is an organism or infectious agent capable of causing disease. A pathogen is not a reproductive hormone: one can infect tissues; the other is a chemical message acting on responsive cells.

Chlamydia, gonorrhea and syphilis are caused by bacteria. HIV, hepatitis B, genital herpes and HPV are caused by viruses. These names identify different infections, not a single disease with one set of symptoms. Some infections are asymptomatic, meaning there are no obvious symptoms. A person’s appearance or lack of discomfort cannot establish that no infection is present. An online symptom list cannot establish a diagnosis either.

Keep the language about biology rather than judging a person’s character, behaviour or identity. The textbook’s historical infection-rate graphs describe particular places, years and age groups. They are not current rates and cannot identify whether an individual has an infection.

Trace how damaged tissue can interrupt a route

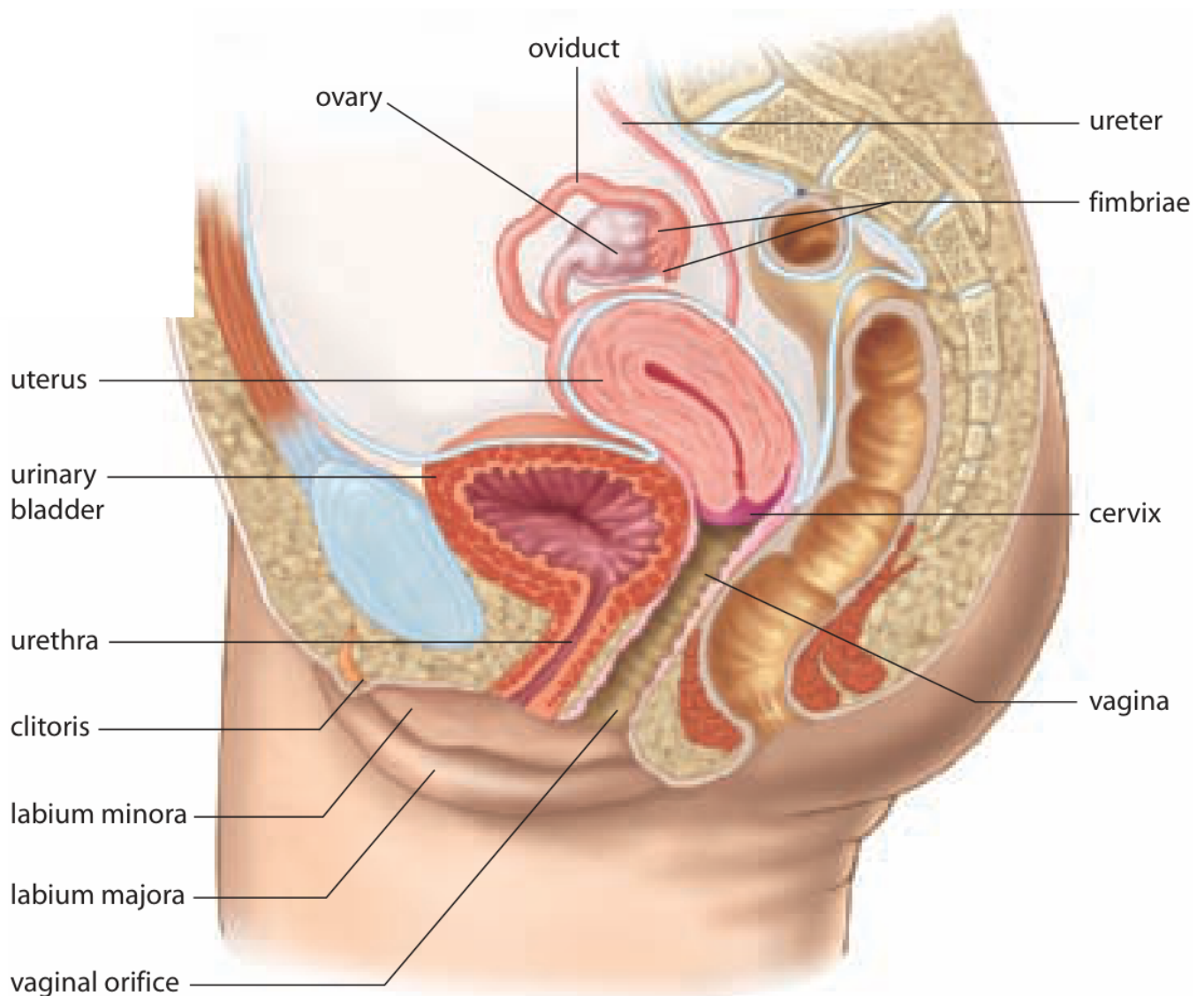
Use the anatomy below to locate an oviduct between the ovary and uterus. There are two oviducts, one associated with each side of the reproductive tract. The diagram shows the normal arrangement; it does not show infection or scarring. Normally a released secondary oocyte enters an oviduct, sperm can travel toward it through the tract, and early development after fertilization moves toward the uterus. Ovulation happens at the ovary; it is not the same event as movement through the duct.

Untreated chlamydia or gonorrhea can spread into upper reproductive structures and contribute to pelvic inflammatory disease, or PID. Inflammation is a tissue response to injury or infection. Damage and healing can leave scar tissue that narrows or blocks an oviduct.

Trace the consequence at that location: sperm and the oocyte may be unable to meet, or an early embryo may not move normally toward the uterus.

This can contribute to infertility, difficulty establishing a pregnancy. It can also increase the risk of an ectopic pregnancy, implantation outside the normal uterine site. Normal fertilization in an oviduct is not itself an ectopic pregnancy; the difference is where the developing embryo implants. Do not conclude that every infected person will develop scarring or infertility.

A similar separation of functions helps in the male tract. The epididymis is the coiled duct where sperm mature and are stored after leaving the testis. Infection and inflammation there can impair sperm function or transport and can affect fertility. Those consequences do not require all testosterone-producing cells to stop functioning. An open route and a functioning hormone system are distinct requirements.



Normal anatomy, reused unchanged from lesson05 and textbook p482. Locate the oviduct; infection and scarring are not drawn here.

Different infections can affect different tissues

Not every STI primarily damages reproductive ducts. HIV affects immune cells, including helper T cells that help coordinate immune responses. Hepatitis B can damage the liver. Some types of human papillomavirus, HPV, can cause cell changes that increase cancer risk; other types cause genital warts. Genital herpes can cause local sores and can also be present when no sores are visible.

Some infections can pass to a fetus or newborn, but the route depends on the pathogen. Syphilis can pass across the placenta during pregnancy. Chlamydia and gonorrhea can pass during birth and cause eye infections; chlamydia can also cause respiratory infection. Herpes can pass during birth and can seriously affect a newborn. Do not turn these examples into a claim that every pathogen crosses the placenta or that every exposure has the same outcome.

Antibiotics act against susceptible bacteria, not viruses. That is why a bacterial treatment cannot simply be transferred to HIV or another viral infection. Bacterial infections can also cause serious lasting harm, so “bacterial” does not mean harmless. Some viral infections have effective antiviral treatment or prevention measures; the details are beyond this mechanism lesson.

Now separate two outcomes: removing an infection and reversing damage it has caused. Treatment can address the pathogen without undoing established scar tissue. This is why the absence of symptoms and the availability of treatment are not evidence that the transport pathway is undamaged. Current professional information, rather than the textbook’s old statistics or treatment statements, is needed for any health decision.

Distinguish preventing conception from reducing transmission

Ask what each method changes. Combined hormonal contraception supplies signals that can suppress the normal LH surge and ovulation. That changes the availability of an oocyte for fertilization. It does not prevent exposure to a pathogen and does not protect against STIs.

A condom acts as a physical barrier. Correct and consistent use of an appropriate condom reduces contact with infectious material and can also obstruct sperm passage. Protection against infection is not absolute: affected skin outside the covered area can still transmit some infections. A barrier does not cure an infection already present, and it does not work by suppressing FSH.

Likewise, interrupting sperm transport through both vasa deferentia is a conception-control mechanism, not a barrier to infection transmission. The fact that sperm cannot complete a route does not mean infectious agents cannot be passed between people. Keep the earlier distinction between sperm cells and accessory-gland fluid in mind. Semen includes sperm and accessory-gland secretions; interrupting the sperm route does not remove all seminal fluid.

Finally, identifying an infection is a separate question from either prevention mechanism. Because symptoms may be absent, appearance alone cannot answer it; testing and appropriate professional care matter. In this lesson, use hypothetical cases and explain the stated biological conditions. No personal sexual or medical information is needed.

Worked example

Treating an infection does not automatically reopen a route

A constructed case states that ovulation and ovarian hormone production continue normally. Earlier infection has left scar tissue that completely blocks both oviducts. Treatment now removes the infection, but the case gives no evidence that the scar tissue or blockage has changed. Can we conclude that the normal route for fertilization has been restored?

- Separate the observations: ovarian activity continues; both transport ducts are blocked; the infection has been removed. None of these statements says the blockage has disappeared.
- Locate the process that can continue: ovulation releases a secondary oocyte at an ovary. Blocking the oviduct does not itself remove that ovarian event.
- Trace the route: the normal passage that allows sperm and the oocyte to meet is obstructed. Normal hormone output does not supply an alternative physical passage.
- Answer the question: no. Removing the pathogen is different from removing established scar tissue. We need separate evidence about the ducts before claiming that the normal route is restored.
- Check the scope: this is an explanation of a specified bilateral blockage, not a diagnosis or a claim that every STI causes infertility.

Optional application

Optional comparison: Model A has open oviducts but no ovulation because the LH surge is suppressed. Model B has normal ovulation but scar tissue blocking both oviducts. Identify the disrupted step in each model. Would removing the infection that caused the old scar tissue necessarily repair either stated problem? Explain.

Answer withheld for first attempt

Practise with support

Apply the explanation

Try both questions. Use the hint to revise your thinking after an incorrect answer.

Why might the absence of symptoms fail to rule out an STI?

- Some infections are asymptomatic
- Only hormone-producing cells can be infected
- All pathogens immediately cause pain

Which consequence most directly follows scarring that blocks an oviduct?

- Automatic loss of all ovarian chromosomes
- Disrupted reproductive-cell transport
- Direct conversion of bacteria into hormones

Stop and think

Try this before opening the explanation

Would treating an infection necessarily remove all scarring it caused?

Answer withheld for first attempt

Open required check · STIs and reproductive function

Explain what you have learned

This check counts toward chapter progress. Correct the 2 selections to unlock 2 written explanations. Saving writing records your response; it does not grade its accuracy.

Selection 1

Which chain best explains one way an STI can reduce fertility?

- Infection → conversion of LH into sperm → duct blockage
- Infection → immediate menstruation in every case → fertilization
- Infection → inflammation and scarring → disrupted transport
- Infection → guaranteed loss of all gametes → higher fertility

Selection 2

Which comparison is accurate?

- No symptoms means no infection
- All contraceptive methods prevent every STI
- Antibiotics treat every viral STI
- Hormonal contraception and STI risk reduction are different functions

Answer the selections correctly to unlock the writing.

Written explanation 1

Explain how an STI could interfere with reproduction while gamete and hormone production continue.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

Written explanation 2

A student says, "If a method prevents pregnancy, it must prevent STIs too." Explain the flaw using two different mechanisms.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

← When hormone patterns change / Next: Chapter review and final check →